IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

INFORMATION THEORY

Friday, 8 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Notation:

(a) The size of a set A is denoted by $|A|$.

(b) By default, the logarithm is to the base 2.

(c) $\oplus$ denotes the exclusive-or operation, or modulo-2 addition.

(d) "i.i.d." means "independent identically distributed".

(e) $H(\cdot)$ is the entropy function.

(f) $C(x) = \frac{1}{2}\log_2(1+x)$ is the capacity function for the Gaussian channel in bits/channel use.

The Questions

1. Basics of information theory.

   a) Suppose x_1 and x_2 are i.i.d. Bernoulli random variables taking values of 0 and 1 with equal probabilities, i.e., $P(x_i = 0) = 0.3$. Let $y = \max(x_1, x_2)$. Compute the following entropy or mutual information terms:

      i) $H(y)$

      ii) $I(x_1 ; y)$

      iii) $I(x_1, x_2; y)$

   [9]

   b) For $\mathbf{p} = [\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}]$ and $\mathbf{q} = [\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$, compute the relative entropy terms $D(\mathbf{p}||\mathbf{q})$ and $D(\mathbf{q}||\mathbf{p})$.

   [8]

   c) x_i is a sequence of i.i.d. Bernoulli random variables with $p(x_i = 1) = p$ where p is unknown. We want to find a function f that converts n samples of x into a smaller number, K, of i.i.d. Bernoulli random variables, z_i, with $p(z_i=1)=½$. Thus $z_{1:K}=f(x_{1:n})$ where K can depend on the values x_i. Show that the following mapping for $n=4$ satisfies the requirements and find the expected value of K.

   0000,1111→ ignore; 1010→0; 0101→1; 0001,0011,0111→00;
   0010,0110,1110→01; 0100,1100,1101→10; 1000,1001,1011→11.

   [8]

2. Typical sequences.

$\mathbf{x}$ and $\mathbf{y}$ are discrete-valued random variables of length n where each pair (x_i, y_i) is drawn independently from the joint probability distribution function $p_{xy}(x,y)$. The jointly typical set $J_\varepsilon^{(n)}$ is the set of vector pairs satisfying the following conditions:

$$J_\varepsilon^{(n)} = \Big\{\mathbf{x},\mathbf{y} : \big|-n^{-1}\log p_x(\mathbf{x}) - H(x)\big| < \varepsilon, \\ \big|-n^{-1}\log p_y(\mathbf{y}) - H(y)\big| < \varepsilon, \\ \big|-n^{-1}\log p_{xy}(\mathbf{x},\mathbf{y}) - H(x,y)\big| < \varepsilon\Big\}$$

where $p_x(x)$ and $p_y(x)$ are the marginal probability distribution functions of x_i and y_i, respectively. The probability $p_x(\mathbf{x}) = \prod_{i=1}^{n} p_x(x_i)$, and similarly for $p_x(\mathbf{x})$ and $p_{xy}(\mathbf{x},\mathbf{y})$.

a) Justify each step in the following derivation of the size of the jointly typical set:

$$(1-\varepsilon)2^{n(H(x,y)-\varepsilon)} \overset{n>N_\varepsilon}{<} \left|J_\varepsilon^{(n)}\right| \le 2^{n(H(x,y)+\varepsilon)}.$$

$$1-\varepsilon \overset{(1)}{<} \sum_{\mathbf{x},\mathbf{y}\in J_\varepsilon^{(n)}} p(\mathbf{x},\mathbf{y}) \overset{(2)}{\le} \left|J_\varepsilon^{(n)}\right| \max_{\mathbf{x},\mathbf{y}\in J_\varepsilon^{(n)}} p(\mathbf{x},\mathbf{y}) \overset{(3)}{\le} \left|J_\varepsilon^{(n)}\right| 2^{-n(H(x,y)-\varepsilon)} \overset{(4)}{\Rightarrow} \left|J_\varepsilon^{(n)}\right| \ge (1-\varepsilon)2^{n(H(x,y)-\varepsilon)}$$

$$1 \overset{(5)}{\ge} \sum_{\mathbf{x},\mathbf{y}\in J_\varepsilon^{(n)}} p(\mathbf{x},\mathbf{y}) \overset{(6)}{\ge} \left|J_\varepsilon^{(n)}\right| \min_{\mathbf{x},\mathbf{y}\in J_\varepsilon^{(n)}} p(\mathbf{x},\mathbf{y}) \overset{(7)}{\ge} \left|J_\varepsilon^{(n)}\right| 2^{-n(H(x,y)+\varepsilon)} \overset{(8)}{\Rightarrow} \left|J_\varepsilon^{(n)}\right| \le 2^{n(H(x,y)+\varepsilon)}$$

[8]

b) Now suppose $n = 9$, $\varepsilon = 0$, and $p_{xy}(x,y)$ given by

$p_{xy}(x,y)$	$y = 0$	$y = 1$
$x = 0$	4/9	1/9
$x = 1$	2/9	2/9

Define the typical set $T_{\mathbf{x}} = \{\mathbf{x} : -n^{-1}\log p_x(\mathbf{x}) = H(x)\}$ for $\varepsilon = 0$.

i) Calculate the probability $P(\mathbf{x} \in T_{\mathbf{x}})$.

ii) Calculate $P(\boldsymbol{x}, \boldsymbol{y} \in J_0^{(9)} \mid \boldsymbol{x} \in T_{\boldsymbol{x}})$.

iii) Determine the value of $P(\boldsymbol{x}, \boldsymbol{y} \in J_0^{(9)})$.

iv) If $\mathbf{z}$ is a random vector, independent of $\mathbf{x}$, whose elements are independent Bernoulli variables with probability $P(z_i = 0) = 2/3$, calculate $P(\boldsymbol{x}, \boldsymbol{z} \in J_0^{(9)})$.

[17]

3. Source and channel coding.

a) Lempel-Ziv coding. Give the LZ78 parsing and encoding of the following sequence:

$$0000001101010010011101$$

[Note: For this question, you will see less than 15 phrases; so ALWAYS use four bits to represent the location of a phrase. Do not worry about how to save such bits.]

[10]

b) Parallel channels and waterfilling. Consider the following three parallel Gaussian channels

$$\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} + \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix}$$

where

$$\begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix} \sim N\left(0, \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}\right)$$

with a power constraint $E(X_1^2 + X_2^2 + X_3^2) \le 3P$. Assume that $\sigma_1^2 \ge \sigma_2^2 \ge \sigma_3^2$. At what power does the channel behave like

i) a single channel with noise variance σ_3^2? Find the channel capacity.

ii) a pair of channels with noise variances σ_3^2 and σ_2^2? Find the channel capacity.

iii) three channels with noise variances σ_3^2, σ_2^2, and σ_1^2? Find the channel capacity.

[15]

4. Network information theory.

a) Consider a two-user multiple access Gaussian channel with reference to Fig. 4.1, where $C(x)$ is the capacity function, N is the noise power, and P_1 and P_2 are the powers of the two users.

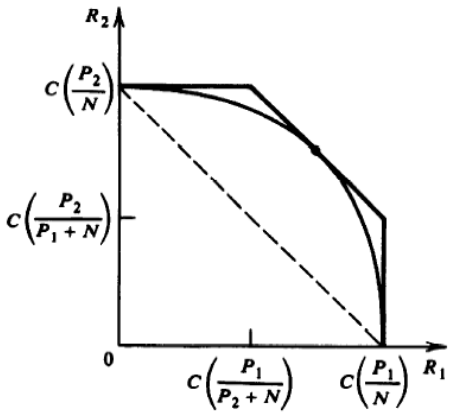


Fig. 4.1. Capacity region of multi-access channel.

The "quarter circle" in Fig. 4.1 is the capacity region of FDMA

$$R_1 \le W_1 C\left(\frac{P_1}{N_0 W_1}\right), \quad R_2 \le W_2 C\left(\frac{P_2}{N_0 W_2}\right)$$

where N_0 is the single-sided power spectral density of the noise.

i) Show that the FDMA region is not larger than the capacity region of the multiple access channel (i.e., CDMA region) with the same bandwidth $W = W_1 + W_2$. (Hint: the function $x\log(1+1/x)$ is increasing and concave for $x > 0$.)

ii) Given the total bandwidth W, find bandwidth W_1 that maximizes the FDMA sum rate $R = R_1 + R_2$.

[15]

b) Find the capacity regions of the following binary multiple-access channels.

i) Additive modulo 2 multiple-access channel where $X_1 \in \{0,1\}$, $X_2 \in \{0,1\}$, $Y = X_1 \oplus X_2$.

ii) Additive multiple-access channel where $X_1 \in \{-1,1\}$, $X_2 \in \{-1,1\}$, $Y = X_1 + X_2$. Here + is addition over the reals.

[10]